Homework 6: Metals

Paper track. No home lab (the metals kit stays at practice). Calculator needed.

Part A: Identify the metal (2 pts each)

#	Color	Magnet	Density (g/cm^3)	With HCl	Metal + symbol + atomic #
1	reddish-orange	no	8.96	no reaction	
2	silvery	no	2.70	slow start, then steady	
3	gray	**yes**	7.87	slow–moderate, solution goes pale green	
4	dull silver	no	1.74	vigorous, fast	
5	bluish-gray	no	7.14	moderate, steady	

Part B: Density (3 pts each)

1. A metal cube 3.0 cm on a side weighs 72.9 g. Density? Which metal?
2. A piece of metal raises water in a graduated cylinder from 25.0 mL to 34.0 mL and weighs 80.6 g. Density? Which metal?
3. Convert 7.14 g/cm^3 to kg/m^3.
4. A solid iron bar has a volume of 50 cm^3. What is its mass? (Fe = 7.87 g/cm^3)
5. A 2.0-meter cube of aluminum — what is its mass in kg? (Al = 2700 kg/m^3)

Part C: Reactions & oxides (2 pts each)

1. Classify: $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$
2. Classify: $Zn + 2\ HCl \rightarrow ZnCl_2 + H_2$
3. Classify: $2\ Mg + O_2 \rightarrow 2\ MgO$
4. Why does aluminum resist acid at first, but iron just keeps corroding?
5. Steel, brass, and bronze — name the metals in each.

Part D: Scenario (3 pts)

A gray metal fragment at a scene is not attracted to a magnet, has a measured density of 2.71 g/cm^3, and reacts slowly with HCl at first and then steadily. Name the metal. A suspect who works with aircraft parts and a suspect who works with cast-iron cookware are both of interest — which does this implicate?

Answer Key

Part A: 1. Copper — Cu — 29. 2. Aluminum — Al — 13. 3. Iron — Fe — 26. 4. Magnesium — Mg — 12. 5. Zinc — Zn — 30.

Part B:

1. Volume = 27.0 cm^3; 72.9 ÷ 27.0 = **2.70 g/cm^3** → **aluminum**.
2. Volume = 9.0 mL; 80.6 ÷ 9.0 ≈ **8.96 g/cm^3** → **copper**.
3. **7140 kg/m^3**.
4. 7.87 × 50 = **393.5 g**.
5. $(2.0)^3$ = 8.0 m^3; 8.0 × 2700 = **21,600 kg**.

Part C:

1. Decomposition. 2. Single replacement. 3. Synthesis (combination).
2. Aluminum forms a thin, tough, **protective** oxide layer (Al_2O_3) that seals the surface (passivation); iron's oxide (rust, $Fe_2O_3 \cdot xH_2O$) is flaky and does not protect, so oxygen keeps reaching fresh metal.
3. Steel = iron + carbon; brass = copper + zinc; bronze = copper + tin.

Part D: Aluminum (non-magnetic, density ~2.70, oxide-layer delay then steady reaction). It implicates the **aircraft-parts** suspect (aircraft use aluminum alloys); the cast-iron-cookware suspect is cleared — that would be iron, which is magnetic and much denser.